technische universität dortmund

PhD Application Presentation

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About Me

Simin Ma

M.Sc. Data Science | TU Dortmund University

Experimental spirit · Collaborative researcher

Background

- M.Sc. Data Science, TU Dortmund
- B.Sc. Applied Statistics
- Study Abroad, University of Malaya

Industry Experience

- **Intrum**:
- Data preprocessing for LLM applications
- Heterogeneous records → standardised text workflow

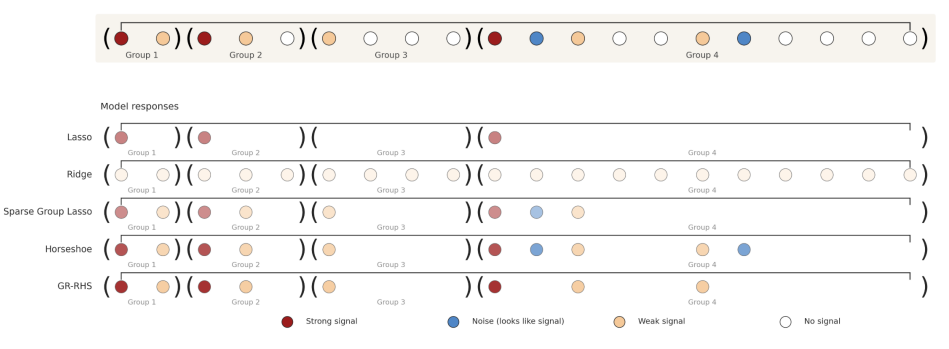
Research Experience

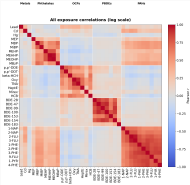
- **TU Dortmund ls 9 Group: UQ in Deep Learning** Experimental Pipeline, reproducible Python/PyTorch workflow
- **TU Dortmund Project: Seminar** GAN Implementation, convergence instability, hyperparameter sensitivity

Master Thesis

Intuitions behind the idea

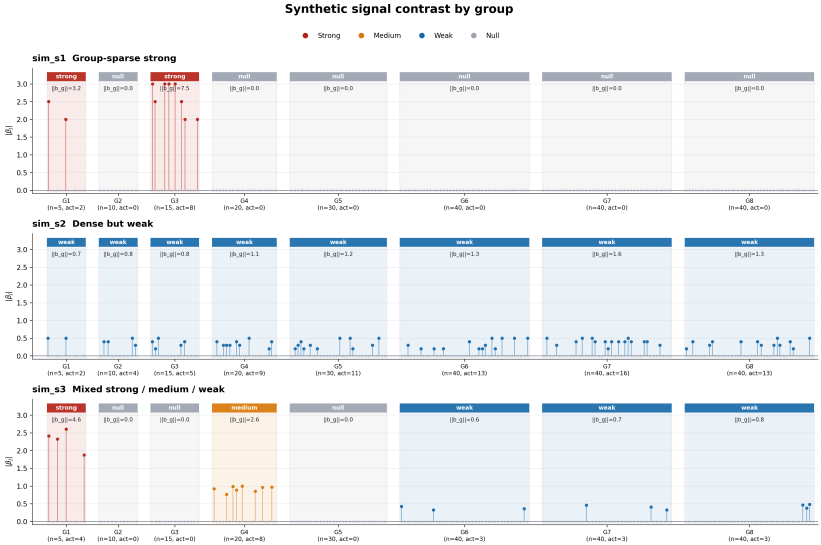
Master Thesis: The Group-Regularized Regularized Horseshoe (GR-RHS) Model for Mixed-Signal Regression.



- 

- Group Structure is normal and natural.

Experiments Design (Simulation Results)



Each panel uses group-wise x positions. Box color marks dominant group type; points show individual |beta_j| values, revealing within-group dense vs sparse signal contrast.

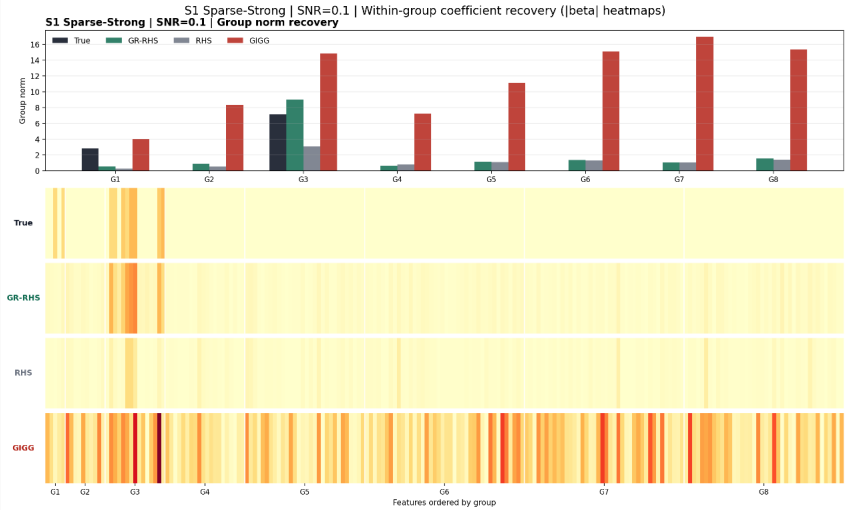
- Shared data & splits
- Fair baseline
- Fixed compute budget
- Prior sensitivity check
- Convergence consistency
- Unified metrics
- Reproducible configs

Experiments Design (Simulation Results)

Table 1: RMSE comparison across scenarios and SNR levels

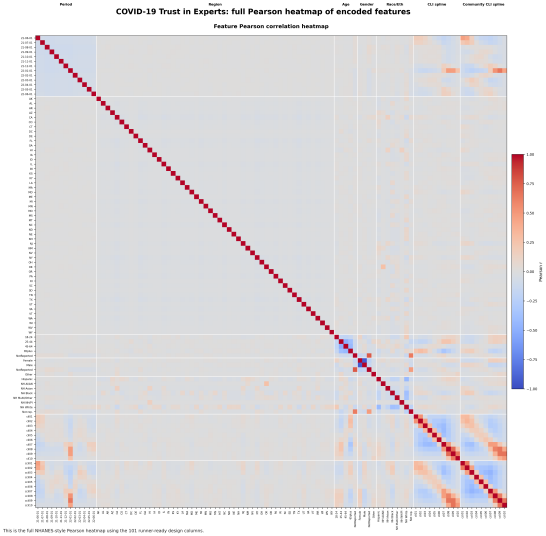
Scenario	SNR	GR-RHS	RHS	GIGG	SGL	Lasso	Ridge
S1	0.1	26.7382	27.1911	40.1707	**24.7224**	24.7534	25.0324
S1	0.5	12.0072	12.5457	17.7489	**11.0136**	11.2779	12.2130
S1	1.0	**7.7705**	8.6790	12.3657	7.8598	7.9754	9.4876
S1	3.0	4.9045	4.9841	6.7307	**4.5205**	4.6106	5.8996
S2	0.1	**10.9243**	10.9953	17.2614	11.2270	11.2453	11.2105
S2	0.5	**5.5266**	5.6159	7.2790	5.5470	5.5844	5.5284
S2	1.0	**4.1304**	4.3570	4.9846	4.1893	4.2769	4.2845
S2	3.0	2.7381	2.7385	2.8270	2.6890	2.7571	**2.6789**
S3	0.1	18.3627	17.2453	29.2454	**16.6957**	16.9668	17.0547
S3	0.5	**7.6227**	7.7513	12.7004	7.6557	7.8260	8.3502
S3	1.0	**5.2247**	5.7138	8.6687	5.5066	5.6145	6.4537
S3	3.0	**3.1485**	3.2738	4.3568	3.3006	3.3057	4.0716

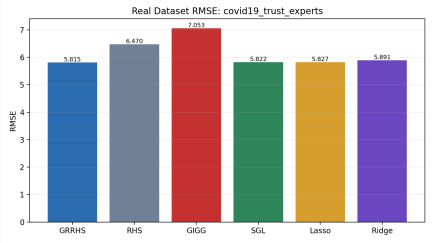
Experiments Design (Simulation Results)



- Do they really recover?

Experiments Design (Real Datasets)





Research Interests

Curious about...

Attention is all you need. But cuda is out-of-memory.

- - Everything is about Trade-Off.
- - Can I trust them?
- - Can it be more user-friendly?

Beautiful things don’t ask for attention.

Thank you for listening and for the opportunity to present!